THEORY REPORT

Assignment No 2

Title: Analyze Wi-Fi Network Information to Identify Trusted Networks and Detect Possible Fake Access Points Using SSID, BSSID, and Security Parameters

1. Aim

To analyze available Wi-Fi network information and identify trusted networks and possible fake or rogue access points using **SSID, BSSID, signal strength, and security parameters**.

2. Objectives

1. To collect information about nearby Wi-Fi networks.
2. To understand the difference between SSID and BSSID.
3. To analyze Wi-Fi security and authentication parameters.
4. To identify suspicious networks that may be fake access points.
5. To recommend suitable measures for safe Wi-Fi connectivity.

3. Introduction

A Wi-Fi network is identified by an **SSID (Service Set Identifier)**, which is the network name displayed to users. Each access point has a unique **BSSID (Basic Service Set Identifier)**, generally represented by the MAC address of the wireless interface.

Attackers can create **fake or rogue access points**, commonly associated with **Evil Twin attacks**, by using the same or a similar SSID as a trusted network. Therefore, checking only the network name is not sufficient. The BSSID, security configuration, signal strength, channel, and vendor information can provide additional indicators for identifying suspicious networks.

4. Wi-Fi Parameters Used for Analysis

SSID:
The visible name of the wireless network. Multiple access points may legitimately use the same SSID, so an identical SSID alone does not prove that a network is fake.

BSSID:
The MAC address identifying a particular access point or wireless interface. Comparing the BSSID with a known trusted network can help distinguish legitimate infrastructure from suspicious devices.

Security:
Indicates the wireless security configuration, such as WPA2-Personal, WPA2-Enterprise, WPA3, WPA, or Open.

Signal Strength:
Measured in dBm. A significantly stronger signal from an unexpected access point using a trusted SSID can be an indicator requiring investigation, although signal strength alone cannot confirm a fake AP.

Channel:
Shows the radio channel used by the access point. Multiple access points can legitimately operate on the same channel.

5. Network Information Collection

Wi-Fi information can be collected using authorized system tools such as Windows Wi-Fi commands or wireless scanning applications.

For example:

netsh wlan show interfaces

This can display the currently connected network, BSSID, channel, authentication method, and signal strength.

Available Wi-Fi profiles can also be examined using:

netsh wlan show profiles

A wireless scanner can provide additional information about nearby SSIDs, BSSIDs, channels, signal levels, and security types.

6. Analysis

The collected information should be compared with the details of known trusted networks.

For example:

SSID	BSSID	Security	Signal	Analysis
Home_Network	Known BSSID	WPA2/WPA3	Strong	Trusted
Office_WiFi	Known BSSID	WPA2-Enterprise	Strong	Trusted
Home_Network	Different BSSID	Open/WPA	Strong	Suspicious
Cafe_Free_WiFi	Unknown BSSID	Open	Weak	Public/Untrusted

A **duplicate SSID with a different BSSID is not automatically a fake access point**, because legitimate networks may use multiple access points or extenders. However, a duplicate SSID combined with unexpected security settings, unknown hardware, or unusual signal behavior should be investigated.

7. Security Risks

Possible fake access points can result in:

- Credential theft
- Eavesdropping
- Man-in-the-Middle attacks
- Phishing
- Session interception

- Malicious redirection
- Exposure of sensitive information

An attacker may make a rogue access point appear attractive by using the same SSID as a legitimate network.

8. Recommended Security Measures

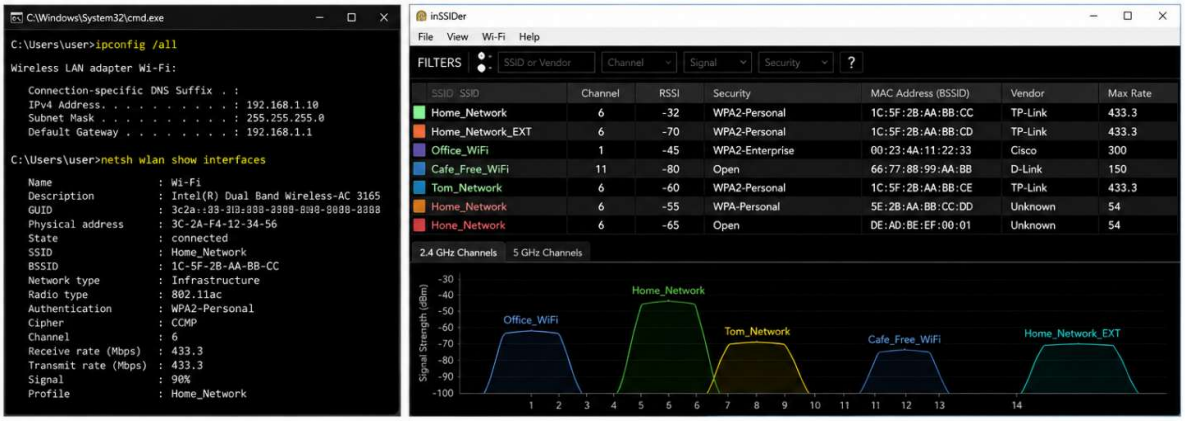
1. Verify the **SSID and BSSID** of trusted networks before connecting where practical.
2. Prefer networks using **WPA2 or WPA3** with appropriate authentication.
3. Avoid unknown or open networks for sensitive activities.
4. Disable automatic connection to unknown Wi-Fi networks.
5. Verify suspicious duplicate SSIDs with the network administrator.
6. Keep router and access-point firmware updated.
7. Use HTTPS/TLS for application and web communication.
8. Use a VPN when operating on an untrusted network where appropriate.
9. Monitor for unexpected access points within organizational environments.

9. Result

Wi-Fi networks were analyzed using SSID, BSSID, signal strength, channel, vendor, and security parameters. Trusted networks could be distinguished from suspicious networks by comparing these characteristics with known network information. Networks with unexpected BSSIDs, weak or open security, and other unusual characteristics were identified as requiring further investigation.

10. Conclusion

SSID alone cannot reliably establish whether a Wi-Fi network is trustworthy. A more effective analysis combines **SSID, BSSID, security configuration, signal strength, channel, and known network information**. Duplicate SSIDs with different BSSIDs should be investigated rather than automatically classified as malicious. Using WPA2/WPA3, avoiding unknown networks, disabling unnecessary auto-connect behavior, and verifying network details can significantly reduce the risk of connecting to fake or rogue access points.


C:\Windows\System32\cmd.exe
C:\Users\user>ipconfig /all
Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . :
IPv4 Address. . . . . : 192.168.1.10
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.1
C:\Users\user>netsh wlan show interfaces
Name : Wi-Fi
Description : Intel(R) Dual Band Wireless-AC 3165
GUID : 3c2a::33-313-333-3333-333-3333-3333
Physical address : 3C-2A-F4-12-34-56
State : connected
SSID : Home_Network
BSSID : 1C-5F-2B-AA-BB-CC
Network type : Infrastructure
Radio type : 802.11ac
Authentication : WPA2-Personal
Cipher : CCMP
Channel : 6
Receive rate (Mbps) : 433.3
Transmit rate (Mbps) : 433.3
Signal : 90%
Profile : Home_Network
inSSIDer
File View Wi-Fi Help
FILTERS
SSID or Vendor
Channel
Signal
Security
SSID SSID
Channel
RSSI
Security
MAC Address (BSSID)
Vendor
Max Rate
Home_Network 6 -32 WPA2-Personal 1C:5F:2B:AA:BB:CC TP-Link 433.3
Home_Network_EXT 6 -70 WPA2-Personal 1C:5F:2B:AA:BB:CD TP-Link 433.3
Office_WiFi 1 -45 WPA2-Enterprise 00:23:4A:11:22:33 Cisco 300
Cafe_Free_WiFi 11 -80 Open 66:77:88:99:AA:BB D-Link 150
Tom_Network 6 -60 WPA2-Personal 1C:5F:2B:AA:BB:CE TP-Link 433.3
Home_Network 6 -55 WPA-Personal 5E:2B:AA:BB:CC:DD Unknown 54
Hone_Network 6 -65 Open DE:AD:BE:EF:00:01 Unknown 54
2.4 GHz Channels
5 GHz Channels
Signal Strength (dBm)
-30
-40
-50
-60
-70
-80
-90
-100
Office_WiFi
Home_Network
Tom_Network
Cafe_Free_WiFi
Home_Network_EXT
1 2 3 4 5 5 6 7 8 9 10 11 11 12 13 14